PowerNext-AI Screening Submission: Methodology

Problem and data. The workbook contains 1000 labeled training rows and 350 unlabeled test rows. The training targets are Reference_Parameter (regression) and Validity_Label (binary classification). Test_ID is retained only for output and is excluded from modeling.

Preprocessing and features. Missing numeric values are median-imputed inside each scikit-learn pipeline, with missingness indicators. Derived features include sensor mean, standard deviation, range, coefficient of variation, all pairwise sensor gaps, a per-row missing-value count, a voltage-current power proxy, and a temperature-duration proxy.

Invalid-record signal engineering. Two leak-free, label-free anomaly features were added before classification: (1) the disagreement between two independently trained regressors (ExtraTrees vs RandomForest) on the predicted Reference_Parameter, computed out-of-fold for training rows and directly for test rows -- rows with inconsistent sensor physics tend to produce disagreeing predictions; (2) an Isolation Forest anomaly score over the full engineered feature space.

Reference prediction. An ExtraTreesRegressor (400 trees, minimum leaf size 2, 90% feature subsampling) was selected using five-fold shuffled cross-validation. Cross-validated MAE was 0.9705, RMSE 2.3535, and R2 0.9515.

Invalid detection. ExtraTreesClassifier and RandomForestClassifier (both with balanced class weighting) were each evaluated with out-of-fold probabilities averaged over 5 independent stratified five-fold splits, using the anomaly-augmented feature set. For each model the decision threshold was chosen to directly maximize out-of-fold invalid-class F1 (not fixed in advance), and the better-performing model, ExtraTreesClassifier, was selected. This produced accuracy 0.9890, invalid precision 0.9624, invalid recall 0.9552, and invalid F1 0.9588 at threshold 0.331.

Validation and assumptions. Regression folds were shuffled K-folds; classification folds were stratified and repeated with different shuffles to stabilize the threshold estimate. Rows are assumed independent (no repeated group key supplied). Extreme values are retained and handled by the ensembles rather than deleted. Reported metrics are validation estimates, not hidden-test performance.

Digital-twin automation. In deployment, equipment sensors feed a timestamped acquisition layer. A validation gateway checks schema, units, ranges, missingness, and duplicate events, then applies the locked preprocessing and feature pipeline (including the disagreement and anomaly-score computations). The digital twin estimates the current operating state, predicts the reference parameter, and assigns a valid/invalid probability. Confidence, drift, and sensor-health monitors route alerts to operators; confirmed outcomes are stored for periodic recalibration, champion/challenger testing, and controlled retraining.

Input features: Applied_Voltage_kV, Load_Current_A, Ambient_Temperature_C, Test_Duration_min, Sensor_S1, Sensor_S2, Sensor_S3, Sensor_S4, Sensor_mean, Sensor_std, Sensor_range, Sensor_cv, Sensor_S1_minus_Sensor_S2, Sensor_S1_minus_Sensor_S3, Sensor_S1_minus_Sensor_S4, Sensor_S2_minus_Sensor_S3, Sensor_S2_minus_Sensor_S4, Sensor_S3_minus_Sensor_S4, Electrical_power_proxy, Temperature_duration_proxy, Missing_count, Reg_disagreement, Isolation_anomaly_score